

UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Relational Algebra

Basi di Dati

Bachelor's Degree in Computer Engineering
Academic Year 2026/2027

Stefano Marchesin

Intelligent Interactive Information Access (IIIA) Hub
Department of Information Engineering
University of Padua

DIPARTIMENTO
DI INGEGNERIA
DELL'INFORMAZIONE



Outline

- Update operation (insertion, modification, and deletion)
- Relational Algebra (query operations)

Update Operations

Operations

- We can classify the relational model operations into
  - **update**
  - **query**

- Given a database schema and a database instance
  - an **update** is a function which maps the given **database instance** into a **new database instance** which is **valid** with respect to all the integrity constraints
  - a **query** is a function which maps the given **database instance** into a **relation**

Insertion

- It may violate
  - domain constraints
  - tuple constraints
  - key constraints
  - referential integrity constraints
- Alternatives
  - **prevent the insertion**
  - try to correct the cause of the constraint violation

Update

- It may violate
  - domain constraints
  - tuple constraints
  - key constraints
  - referential integrity constraints
- Alternatives
  - prevent the update (***restrict***)
  - propagate the update (***cascade***)

© Stefano Marchesin

Deletion

- It may violate
  - referential integrity constraints
- Alternatives
  - prevent the deletion (***restrict***)
  - propagate the deletion (***cascade***)
  - modify the values of the referenced attributes setting them to **NULL** (***set null***) or to a default value (***set default***)
    - note that, if the referenced attributes are part of the primary key, it is not possible to set them to NULL

© Stefano Marchesin

Relational Algebra (query operations)

Relational Algebra

The relational algebra is a set of operators to manipulate whole relations: each operator processes one (or more) relations and produces as output a new relation

- Relational algebra operators can be **composed** to create **expressions**
- The relational algebra is a **procedural language**

© Stefano Marchesin

Relational Algebra Operators

- Primitive operators
  - $\cup$ union
  - $-$ difference
  - $\times$ product
  - ρ rename
  - σ selection
  - π projection
- Derived operators
  - $\cap$ intersection
  - $\bowtie$ natural join
  - $\bowtie_\Theta$ theta join
  - $\overset{\leftarrow}{\bowtie}_\Theta$ $\overset{\rightarrow}{\bowtie}_\Theta$ $\overset{\leftrightarrow}{\bowtie}_\Theta$ outer join
- Set operators
  - $\cup$ union
  - $-$ difference
  - $\times$ product
  - $\cap$ intersection
- Relational operators
  - ρ rename
  - σ selection
  - π projection
  - $\bowtie$ natural join
  - $\bowtie_\Theta$ theta join
  - $\overset{\leftarrow}{\bowtie}_\Theta$ $\overset{\rightarrow}{\bowtie}_\Theta$ $\overset{\leftrightarrow}{\bowtie}_\Theta$ outer join

Compatibility to Union

Two relations $R_1(X_1, X_2, \ldots, X_n)$ and $R_2(Y_1, Y_2, \ldots, Y_n)$ are compatible to union if they have the same degree n and

$$\begin{cases} X_i = Y_i \\ \mathrm{dom}(X_i) = \mathrm{dom}(Y_i) \end{cases} \qquad 1 \leq i \leq n$$

- Two relations are compatible to union when they have the same number of attributes and each pair of attributes has the same name and domain
- For the intersection, union, and difference operators, the two involved relations must be compatible to union

Intersection

Given two relations $R_1(X)$ and $R_2(Y)$ compatible to union, their **intersection** is the relation

$$R_1 \cap R_2 = \{t \mid t \in R_1 \wedge t \in R_2\}$$

Graduated

Badge	Surname	Age
7274	Rossi	42
7432	Neri	54
9824	Verdi	45

Manager

Badge	Surname	Age
9297	Neri	33
7432	Neri	54
9824	Verdi	45

Graduated ∩ Manager

Badge	Surname	Age
7432	Neri	54
9824	Verdi	45

Union

Given two relations $R_1(X)$ and $R_2(Y)$ compatible to union, their **union** is the relation

$$R_1 \cup R_2 = \{t \mid t \in R_1 \vee t \in R_2\}$$

Graduated

MatriBadg	Surname	Age
7274	Rossi	42
7432	Neri	54
9824	Verdi	45

Manager

Badge	Surname	Age
9297	Neri	33
7432	Neri	54
9824	Verdi	45

Graduated ∪ Manager

Badge	Surname	Age
7274	Rossi	42
7432	Neri	54
9824	Verdi	45
9297	Neri	33

Difference

Given two relations $R_1(X)$ and $R_2(Y)$ compatible to union, their difference is the relation

$$R_1 - R_2 = \{t \mid t \in R_1 \wedge t \notin R_2\}$$

Graduated

Badge	Surname	Age
7274	Rossi	42
7432	Neri	54
9824	Verdi	45

Manager

Badge	Surname	Age
9297	Neri	33
7432	Neri	54
9824	Verdi	45

Graduated − Manager

Badge	Surname	Age
7274	Rossi	42

© Stefano Marchesin

Product (cartesian)

Given two relations $R_1(X)$ and $R_2(Y)$, their product (cartesian) is the relation

$$R_1 \times R_2 = \{xy \mid x \in R_1 \wedge y \in R_2\}$$

- Given the two relations $R_1(X_1, X_2, \ldots, X_n)$ and $R_2(Y_1, Y_2, \ldots, Y_m)$, the cartesian product

$$Q = R_1 \times R_2 = (X_1, X_2, \ldots, X_n, Y_1, Y_2, \ldots, Y_m)$$

- has degree $q = n + m$
- Let $|R_1| = n_{R_1}$ and $|R_2| = n_{R_2}$ be the cardinalities of the two relations, then the cardinality of the cartesian product is

$$|Q| = n_{R_1} * n_{R_2} = n_Q$$

- Note that the use of the term “cartesian product” is somehow improper because the result of this operation is not a pair of tuples but the set of tuples obtained concatenating each tuple of the first relation with each tuple of the second relation

Product (cartesian): example

Product (cartesian): example

Graduated

Badge	Surname	Age
7274	Rossi	42
7432	Neri	54
9824	Verdi	45

Manager

Badge	Surname	Age
9297	Neri	33
7432	Neri	54
9824	Verdi	45

Graduated × Manager

Badge	Surname	Age	Badge	Surname	Age

Product (cartesian): example

Graduated

Badge	Surname	Age
7274	Rossi	42
7432	Neri	54
9824	Verdi	45

Manager

Badge	Surname	Age
9297	Neri	33
7432	Neri	54
9824	Verdi	45

Graduated × Manager

Badge	Surname	Age	Badge	Surname	Age
7274	Rossi	42	9297	Neri	33
7274	Rossi	42	7432	Neri	54
7274	Rossi	42	9824	Verdi	45
7432	Neri	54	9297	Neri	33
7432	Neri	54	7432	Neri	54
7432	Neri	54	9824	Verdi	45
9824	Verdi	45	9297	Neri	33
9824	Verdi	45	7432	Neri	54
9824	Verdi	45	9824	Verdi	45

© Stefano Marchesin

Rename

Given a relation $R(X)$ and two sets of attributes $A_1, A_2, \ldots, A_m \in X$ and $B_1, B_2, \ldots, B_m \notin X$, the rename is the relation with attributes $X - \{A_1, A_2, \ldots, A_m\} \cup \{B_1, B_2, \ldots, B_m\}$ such that

$$\rho_{B_1,B_2,\ldots,B_m \leftarrow A_1,A_2,\ldots,A_m}(R) = \{t \mid \text{exists } x \in R \text{ such that } t[B_1, B_2, \ldots, B_m] = x[A_1, A_2, \ldots, A_m] \wedge t[C] = x[C] \text{ if } C \neq A_1, A_2, \ldots, A_m\}$$

- This definition states that we are changing only the attribute names but not their domain or the associated values
  - we are modifying the schema but not the instance

Rename: Example (1/2)

Paternity

Father	Child
Adamo	Abele
Adamo	Caino
Abramo	Isacco

Maternity

Mother	Child
Eva	Abele
Eva	Set
Sara	Isacco

Paternity ∪ Maternity ?

Rename: Example (1/2)

$\rho_{Parent \leftarrow Father}(Paternity)$

Parent	Child
Adamo	Abele
Adamo	Caino
Abramo	Isacco

Maternity

Mother	Child
Eva	Abele
Eva	Set
Sara	Isacco

Rename: Example (1/2)

$\rho_{Parent \leftarrow Father}(Paternity)$

Parent	Child
Adamo	Abele
Adamo	Caino
Abramo	Isacco

$\rho_{Parent \leftarrow Mother}(Maternity)$

Parent	Child
Eva	Abele
Eva	Set
Sara	Isacco

Rename: Example (1/2)

$\rho_{Parent \leftarrow Father}(Paternity)$

Parent	Child
Adamo	Abele
Adamo	Caino
Abramo	Isacco

$\rho_{Parent \leftarrow Mother}(Maternity)$

Parent	Child
Eva	Abele
Eva	Set
Sara	Isacco

$\rho_{Parent \leftarrow Father}(Paternity) \cup \rho_{Parent \leftarrow Mother}(Maternity)$

Parent	Child
Adamo	Abele
Adamo	Caino
Abramo	Isacco
Eva	Abele
Eva	Set
Sara	Isacco

Rename: Example (2/2)

Clerk

Surname	Office	Salary
Rossi	Roma	55
Neri	Milano	64

Worker

Surname	Factory	Wages
Bruni	Monza	45
Verdi	Latina	55

$\rho_{Site,\ Pay \leftarrow Office,\ Salary}(Clerk)$

$\cup$

$\rho_{Site,\ Pay \leftarrow Factory,\ Wages}(Worker)$

Surname	Site	Pay
Rossi	Roma	55
Neri	Milano	64
Bruni	Monza	45
Verdi	Latina	55

Selection

Given a relation $R(X)$ and a proposition Θ, the selection is the relation:

$$\sigma_{\Theta}(R) = \{t \mid t \in R \wedge \Theta\}$$

- The selection produces a result which
  - has the same schema as the operand
  - contains a subset of the tuples of the operand, i.e. those tuples which match the condition expressed by the proposition
- The proposition Θ can be defined as follows
  - $X_i\ \theta\ X_j$ with X_i and X_j attributes of R on the same domain and $\theta \in \{<, >, =, \neq, \leq, \geq\}$ comparison operator
  - $X_i\ \theta\ c$ with X_i attribute of R, θ comparison operator and $c \in \mathrm{dom}(X_i)$ constant
  - if ϕ and ψ are propositions, then also $\phi \wedge \psi,\ \phi \vee \psi,\ \neg\phi$ are propositions

Selection: Exercise (1/3)

Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

Find the employees who earn more than 50



Selection: Exercise (1/3)

Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

Find the employees who earn more than 50



$\sigma_{Salary>50}(Employee)$

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
5698	Rossi	Roma	64

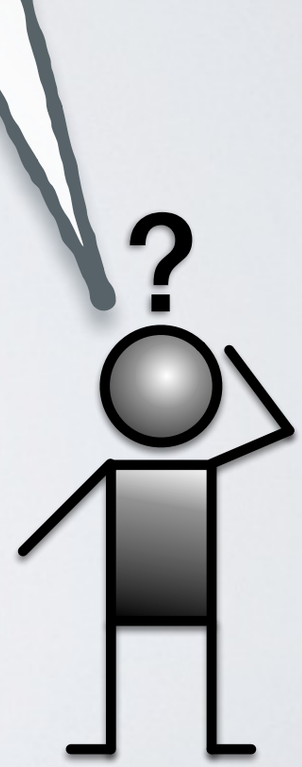
© Stefano Marchesin

Selection: Exercise (2/3)

Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

Find the employees who earn more than 50 and work in Milan



Selection: Exercise (2/3)

Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

Find the employees who earn more than 50 and work in Milan



$\sigma_{Salary>50 \wedge Branch = Milano}$ (Employee)

Badge	Surname	Branch	Salary
5998	Neri	Milano	64

Selection: Exercise (3/3)

Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

Find the employees who have the same surname as the branch where they work

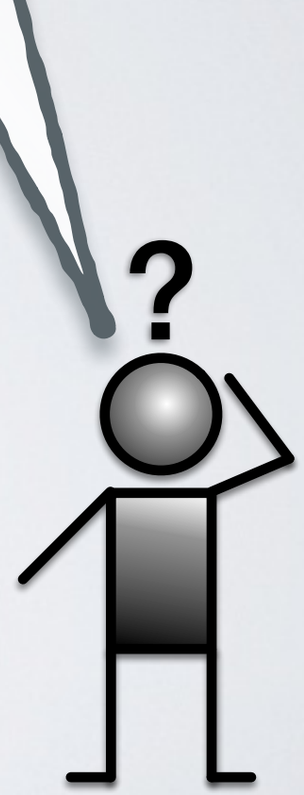


Selection: Exercise (3/3)

Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

Find the employees who have the same surname as the branch where they work



$\sigma_{Surname = Branch}$ (Employee)

Badge	Surname	Branch	Salary
9553	Milano	Milano	44

© Stefano Marchesin

Projection

Given a relation $R(X)$ and a list of its attributes $A_1, A_2, \ldots, A_m$, the projection is the relation:

$$\pi_{A_1,A_2,\ldots,A_m}(R) = \{t[A_1, A_2, \ldots, A_m] \mid t \in R\}$$

- The projection produces a result which
  - has a subset of the attributes of the operand
  - contains (a subset of) the tuples of the operand
- The degree of a projection is m
- The cardinality of a projection is
  - less than or equal to the cardinality of the input relation; less than if the projection contains duplicated tuples which are removed because of the definition of relation
  - equal to the cardinality of the input relation if the attributes $A_1, A_2, \ldots, A_m$ are a super-key

Projection: Exercise (1/2)

Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

Find the badge number and the surname of all the employees



Projection: Exercise (1/2)

Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

Find the badge number and the surname of all the employees



$\pi_{Badge,\ Surname}(Employee)$

Badge	Surname
7309	Rossi
5998	Neri
9553	Milano
5698	Rossi

Projection: Exercise (2/2)

Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

Find surname and branch of all the employees



Projection: Exercise (2/2)

Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

Find surname and branch of all the employees



$\pi_{Surname, Branch}$(Employee)

Surname	Branch
Rossi	Roma
Neri	Milano
Milano	Milano

Selection and Projection

Employee

Badge	Surnamy	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

- Selection and projection are two “orthogonal”, i.e. complementary operators
- Selection performs a **horizontal decomposition**
- Projection performs a **vertical decomposition**

Selection and Projection

Employee

Badge	Surnamy	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

σ

- Selection and projection are two “orthogonal”, i.e. complementary operators
- Selection performs a **horizontal decomposition**
- Projection performs a **vertical decomposition**

Selection and Projection

Employee

Badge	Surnamy	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

σ

π

- Selection and projection are two “orthogonal”, i.e. complementary operators
- Selection performs a **horizontal decomposition**
- Projection performs a **vertical decomposition**

Selection and Projection: Exercise

Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

Find badge number and surname of all the employees who earn more than 50



Selection and Projection: Exercise

Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

Find badge number and surname of all the employees who earn more than 50



$\pi_{Badge,\ Surname}(\sigma_{Salary>50}(Employee))$

Badge	Surname
7309	Rossi
5998	Neri
5698	Rossi

Selection and Projection: Exercise

Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

Find badge number and surname of all the employees who earn more than 50



$\pi_{Badge,\ Surname}(\sigma_{Salary>50}(Employee))$

=

$\sigma_{Sakary>50}(\pi_{Badge,\ Surname}(Employee))$

Selection and Projection: Exercise

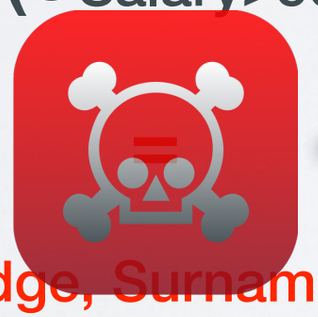
Employee

Badge	Surname	Branch	Salary
7309	Rossi	Roma	55
5998	Neri	Milano	64
9553	Milano	Milano	44
5698	Rossi	Roma	64

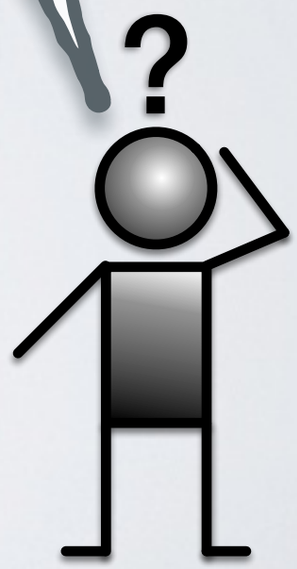
Find badge number and surname of all the employees who earn more than 50

$\pi_{Badge, Surname}(\sigma_{Salary>50}(Employee))$

In general, selection and projection are **not commutative**. They are commutative only when the attributes of the selection are a subset of the attributes of the projection



$\sigma_{Sakary>50}(\pi_{Badge, Surname}(Employee))$



Theta-join

Given two relations $R_1(X)$ and $R_2(Y)$and a proposition Θ, the theta-join is the relation

$$R_1 \bowtie_\Theta R_2 = \{t \mid \text{exists } x \in R_1, y \in R_2 \text{ such that } t[X] = x \wedge t[Y] = y \wedge \Theta\}$$

- Given two relations $R_1(X_1, X_2, \ldots, X_n)$ and $R_2(Y_1, Y_2, \ldots, Y_m)$, the theta-join
$$Q = R_1 \bowtie_\Theta R_2 = (X_1, X_2, \ldots, X_n, Y_1, Y_2, \ldots, Y_m)$$
has degree $q = n + m$
- Let $|R_1| = n_{R_1}$ and $|R_2| = n_{R_2}$ be the cardinalities of the two relations, the cardinality of the theta-join, also called **selectivity**, is
$$0 \leq |Q| = n_Q \leq n_{R_1} * n_{R_2}$$
- The proposition Θ can be defined as follows
  - $X_i \; \theta \; Y_i$ with X_i and Y_i attributes of R_1 and R_2 on the same domain and $\theta \in \{<, >, =, \neq, \leq, \geq\}$ comparison operator
  - if ϕ and ψ are propositions, then also $\phi \wedge \psi$ is a proposition

Theta-join

Given two relations $R_1(X)$ and $R_2(Y)$ and a proposition Θ, the theta-join is the relation

$$R_1 \bowtie_\Theta R_2 = \{t \mid \text{exists } x \in R_1, y \in R_2 \text{ such that } t[X] = x \wedge t[Y] = y \wedge \Theta\}$$

Equivalence

$$R_1 \bowtie_\Theta R_2 = \sigma_\Theta(R_1 \times R_2)$$

(non procedural)

- Given two relations $R_1(X_1, X_2, \ldots, X_n)$ and $R_2(Y_1, \ldots, Y_m)$, the theta-join $Q = R_1 \bowtie_\Theta R_2$ has degree $q = n + m$
- Let $|R_1| = n_{R_1}$ and $|R_2| = n_{R_2}$ be the cardinalities of the two relations, the cardinality of the theta-join, also called **selectivity**, is $0 \leq |Q| = n_Q \leq n_{R_1} * n_{R_2}$
- The proposition Θ can be defined as follows
  - $X_i \; \theta \; Y_i$ with X_i and Y_i attributes of R_1 and R_2 on the same domain and $\theta \in \{<, >, =, \neq, \leq, \geq\}$ comparison operator
  - if ϕ and ψ are propositions, then also $\phi \wedge \psi$ is a proposition

Equi-Join

The **equi-join** is a **theta-join** where we can use only the **=** comparison operator

Equi-Join

The **equi-join** is a **theta-join** where we can use only the **=** comparison operator

Employee

Surname	Dep
Rossi	A
Neri	B
Bianchi	B

Department

Code	Manager
A	Mori
B	Bruni

Employee $\bowtie_{\text{Dep = Code}}$ **Department**

Surname	Dep	Code	Manager

Equi-Join

The equi-join is a theta-join where we can use only the = comparison operator

Employee

Surname	Dep
Rossi	A
Neri	B
Bianchi	B

Department

Code	Manager
A	Mori
B	Bruni

Employee $\bowtie_{Dep = Code}$ Department

Surname	Dep	Code	Manager
Rossi	A	A	Mori
Neri	B	B	Bruni
Bianchi	B	B	Bruni

© Stefano Marchesin

Natural Join

The **natural join** is an **equi-join** where the **attributes** of the two relations in the join condition have the **same names** and the **duplicated attributes** are **removed** from the schema of the output relation

- Let $R_1(XW)$ and $R_2(YZ)$ the two relations with W and Z sets of attributes with the same name, then the natural join can be expressed as

$$R_1 \bowtie R_2 = \pi_{X,W_1,W_2,\ldots,W_m,Y}\left(R_1 \bowtie_{W_1=Z_1 \wedge W_2=Z_2 \wedge \ldots \wedge W_m=Z_m} R_2\right)$$

Natural Join: Example

Natural Join: Example

Employee

Surname	Dep
Rossi	A
Neri	B
Bianchi	B

Department

Dep	Manager
A	Mori
B	Bruni

Employee ⋈ Department

Surname	Dep	Manager

Natural Join: Example

Employee

Surname	Dep
Rossi	A
Neri	B
Bianchi	B

Department

Dep	Manager
A	Mori
B	Bruni

Employee ⋈ Department

Surname	Dep	Manager
Rossi	A	Mori
Neri	B	Bruni
Bianchi	B	Bruni

Cardinality of the Natural Join: Example (1/3)

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Dep	Manager
A	Mori
B	Bruni
B	Verdi

Employee ⋈ Department

Surname	Dep	Manager
Rossi	A	Mori
Neri	B	Bruni
Neri	B	Verdi
Bianchi	B	Bruni
Bianchi	B	Verdi

Cardinality of the Natural Join: Example (1/3)

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Dep	Manager
A	Mori
B	Bruni
C	Verdi

Dep is not a key for R_2

$$|R_1 \bowtie R_2| = 5 \leq |R_1| * |R_2| = 12$$

Employee ⋈ Department

Surname	Dep	Manager
Rossi	A	Mori
Neri	B	Bruni
Neri	B	Verdi
Bianchi	B	Bruni
Bianchi	B	Verdi

Cardinality of the Natural Join: Example (2/3)

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Dep	Manager
A	Mori
B	Bruni
D	Verdi

Employee ⋈ Department

Surname	Dep	Manager
Rossi	A	Mori
Neri	B	Bruni
Bianchi	B	Bruni

Cardinality of the Natural Join: Example (2/3)

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Dep	Manager
A	Mori
B	Bruni
D	Verdi

Dep is a key (primary) for R_2 but it is not a foreign key for R_1

$$|R_1 \bowtie R_2| = 3 \leq |R_1| = 4$$

Employee ⋈ Department

Surname	Dep	Manager
Rossi	A	Mori
Neri	B	Bruni
Bianchi	B	Bruni

Cardinality of the Natural Join: Example (3/3)

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	D

Department (R_2)

Dep	Manager
A	Mori
B	Bruni
D	Verdi

Employee ⋈ Department

Surname	Rep	Manager
Rossi	A	Mori
Neri	B	Bruni
Bianchi	B	Bruni
Neri	D	Verdi

Cardinality of the Natural Join: Example (3/3)

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	D

Department (R_2)

Dep	Manager
A	Mori
B	Bruni
D	Verdi

Dep is a key (primary) for R_2 and a foreign key for R_1

$$|R_1 \bowtie R_2| = 4 = |R_1| = 4$$

Employee ⋈ Department

Surname	Dep	Manager
Rossi	A	Mori
Neri	B	Bruni
Bianchi	B	Bruni
Neri	D	Verdi

Cardinality of the Natural Join (Equi-Join)

- As in the general case, the cardinality of the natural join (equi-join) is

$$0 \leq |R_1 \bowtie R_2| \leq |R_1| * |R_2|$$

- If the natural join (equi-join) involves attributes K which are a key (maybe primary, PK) of R_2 then

$$0 \leq |R_1 \bowtie R_2| \leq |R_1|$$

- If the natural join (equi-join) involves attributes K which are a key (maybe primary, PK) of R_2 and the same attributes are a foreign key FK of R_1 with a referential integrity constraint on R_2 then

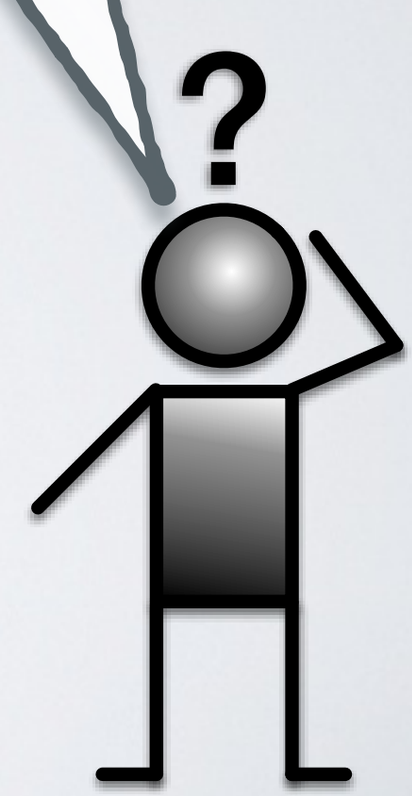
$$|R_1 \bowtie R_2| = |R_1|$$

Relational Algebra: Exercise (1/8)

Employee	Manager
7309	5698
5998	5698
9553	4076
5698	4076
4076	8123

Manage

Find badge, surname, age and salary of the employees who earn more than 42

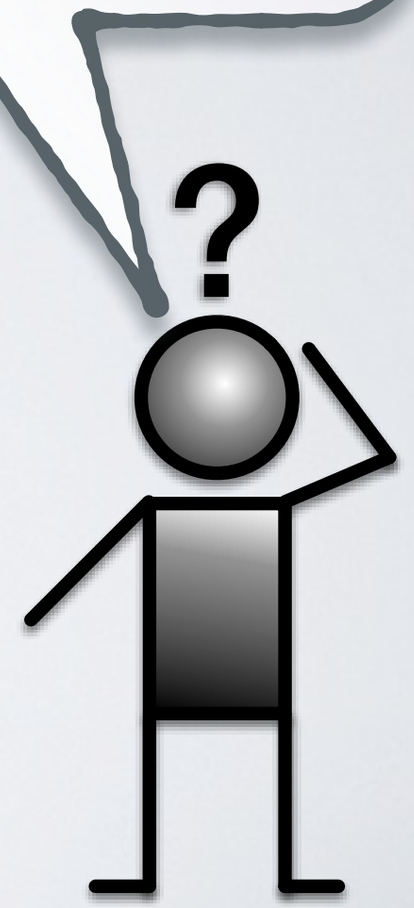


Employee

Badge	Surname	Age	Salary
7309	Rossi	34	45
5998	Bianchi	37	38
9553	Neri	42	35
5698	Bruni	43	42
4076	Mori	45	50
8123	Lupi	46	60

Relational Algebra: Exercise (1/8)

Find badge, surname, age and salary of the employees who earn more than 42



$\sigma_{\text{Salary}>42}(\text{Employee})$

Badge	Surname	Age	Salary
7309	Rossi	34	45
4076	Mori	45	50
8123	Lupi	46	60

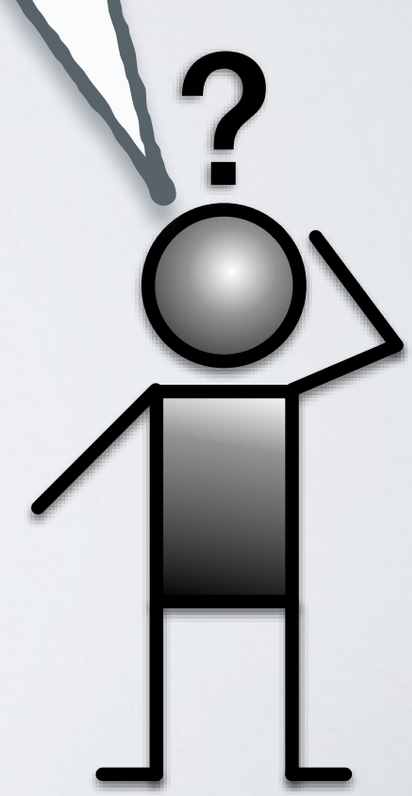
© Stefano Marchesin

Relational Algebra: Exercise (2/8)

Employee	Manager
7309	5698
5998	5698
9553	4076
5698	4076
4076	8123

Manage

Find badge, surname, and age of the employees who earn more than 42



Employee

Badge	Surname	Age	Salary
7309	Rossi	34	45
5998	Bianchi	37	38
9553	Neri	42	35
5698	Bruni	43	42
4076	Mori	45	50
8123	Lupi	46	60

Relational Algebra: Exercise (2/8)

Find badge, surname, and age of the employees who earn more than 42

$$\pi_{\text{Badge,Surname,Age}}(\sigma_{\text{Salary}>42}(\text{Employee}))$$

Badge	Surname	Age
7309	Rossi	34
4076	Mori	45
8123	Lupi	46

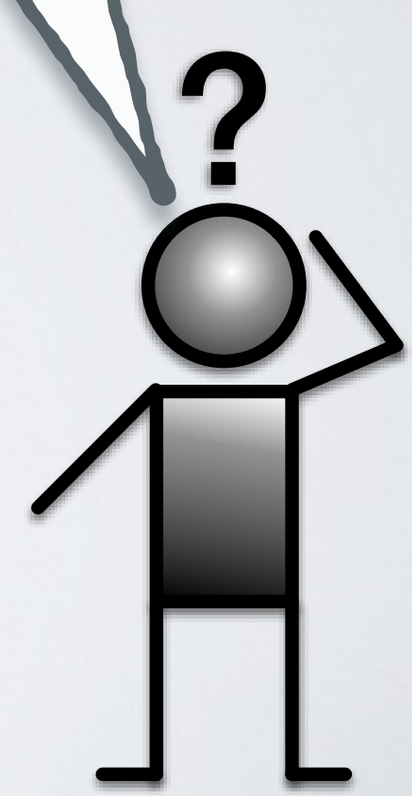


Relational Algebra: Exercise (3/8)

Employee	Manager
7309	5698
5998	5698
9553	4076
5698	4076
4076	8123

Manage

Find the badge number of the managers whose employees earn more than 42



Employee

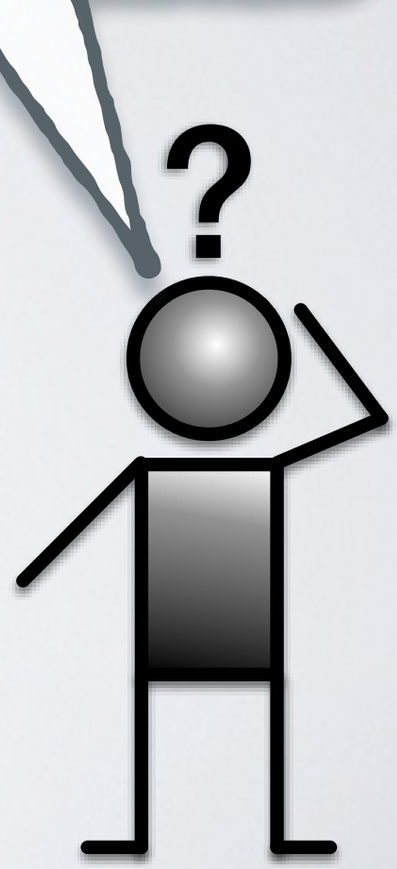
Badge	Surname	Age	Salary
7309	Rossi	34	45
5998	Bianchi	37	38
9553	Neri	42	35
5698	Bruni	43	42
4076	Mori	45	50
8123	Lupi	46	60

Relational Algebra: Exercise (3/8)

Find the badge number of the managers whose employees earn more than 42

$$\pi_{\text{Manager}}(\text{Manage} \bowtie_{\text{Employee}=\text{Badge}} \sigma_{\text{Salary}>42}(\text{Employee}))$$

Manager
5698
8123



Relational Algebra: Exercise (4/8)

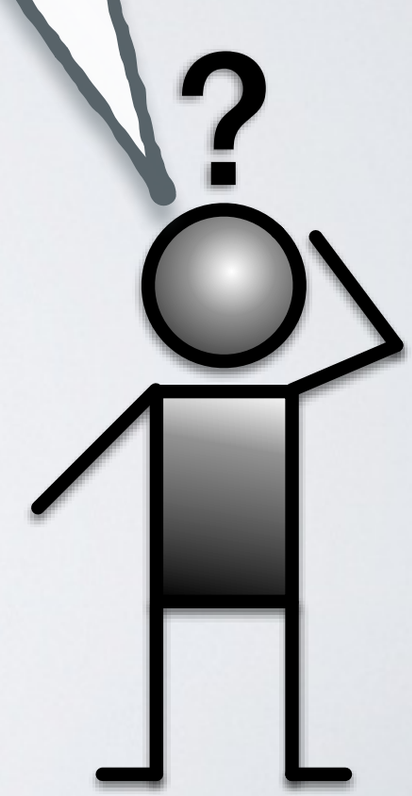
Manage

Employee	Manager
7309	5698
5998	5698
9553	4076
5698	4076
4076	8123

Employee

Badge	Surname	Age	Salary
7309	Rossi	34	45
5998	Bianchi	37	38
9553	Neri	42	35
5698	Bruni	43	42
4076	Mori	45	50
8123	Lupi	46	60

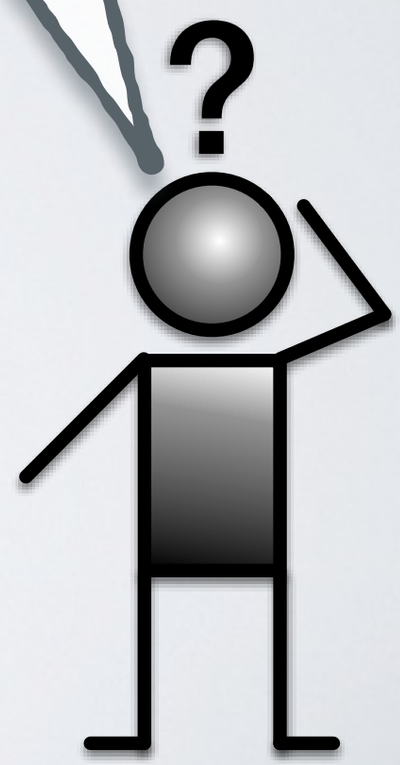
Find the surname and the salary of the managers whose employees earn more than 42



Relational Algebra: Exercise (4/8)

Find the surname and the salary of the managers whose employees earn more than 42

$$\pi_{\text{Surname,Salary}}\Big(\text{Employee} \bowtie_{\text{Badge}=\text{Manager}} \text{Manage} \bowtie_{\text{Employee}=\text{E.Bdg}} \rho_{\text{E.Bdg,E.Srn,E.Age,E.Slry}\leftarrow\text{Badge,Surname,Age,Salary}}\big(\sigma_{\text{Salary}>42}(\text{Employee})\big)\Big)$$



Surname	Salary
Bruni	42
Lupi	60

Relational Algebra: Exercise (4/8)

$$\pi_{\text{Surname,Salary}}\Big(\text{Employee} \bowtie_{\text{Badge}=\text{Manager}} \pi_{\text{Manager}}\big(\text{Manage} \bowtie_{\text{Employee}=\text{Badge}} (\sigma_{\text{Salary}>42}(\text{Employee}))\big)\Big)$$

Find the surname and the salary of the managers whose employees earn more than 42



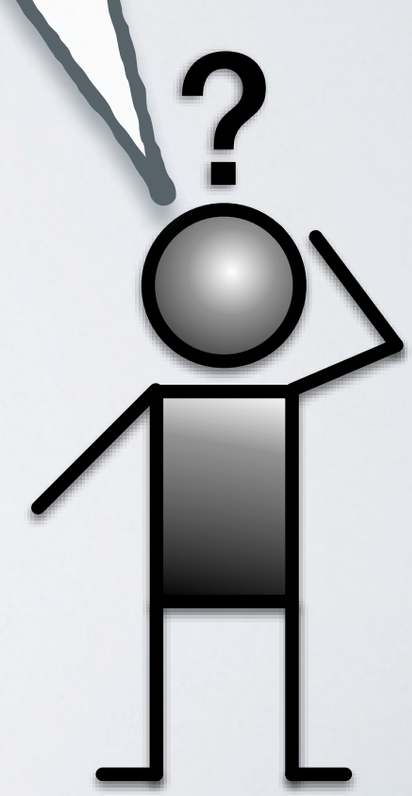
Surname	Salary
Bruni	42
Lupi	60

Relational Algebra: Exercise (5/8)

Employee	Manager
7309	5698
5998	5698
9553	4076
5698	4076
4076	8123

Manage

Find the surname and the salary of the managers who earn more than 45



Employee

Badge	Surname	Age	Salary
7309	Rossi	34	45
5998	Bianchi	37	38
9553	Neri	42	35
5698	Bruni	43	42
4076	Mori	45	50
8123	Lupi	46	60

© Stefano Marchesin

Relational Algebra: Exercise (5/8)

Find the surname and the salary of the managers who earn more than 45

$$\pi_{\text{Surname,Salary}}(\text{Manage} \bowtie_{\text{Manager}=\text{Badge}} \sigma_{\text{Salary}>45}(\text{Employee}))$$

Surname	Salary
Mori	50
Lupi	60



Relational Algebra: Exercise (6/8)

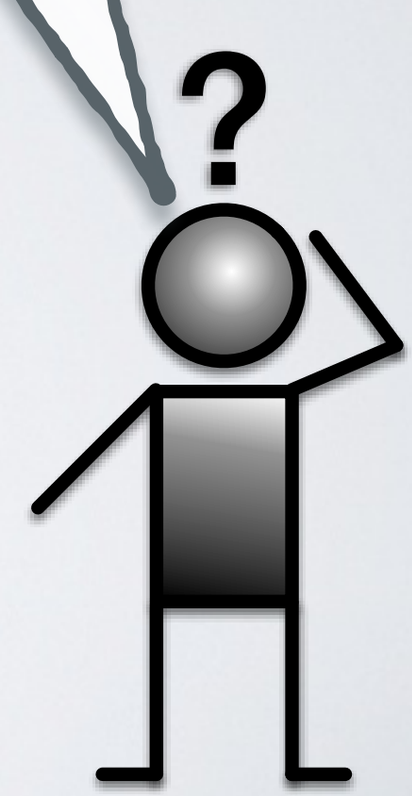
Employee	Manager
7309	5698
5998	5698
9553	4076
5698	4076
4076	8123

Manage

Employee

Badge	Surname	Age	Salary
7309	Rossi	34	45
5998	Bianchi	37	38
9553	Neri	42	35
5698	Bruni	43	42
4076	Mori	45	50
8123	Lupi	46	60

Find the employees who earn more than their managers, returning badge, surname and salary of both the employee and the manager



Relational Algebra: Exercise (6/8)

$$\pi_{\text{Badge,Surname,Salary,M.Bdg,M.Srn,M.Slry}}\Big($$

$$\sigma_{\text{Salary}>\text{M.Slry}}\Big($$

$$\rho_{\text{M.Bdg,M.Srn,M.Age,M.Slry}\leftarrow\text{Badge,Surname,Age,Salary}}(\text{Employee})$$

$$\bowtie_{\text{M.Bdg}=\text{Manager}}$$

$$\text{Manager} \bowtie_{\text{Employee}=\text{Badge}} \text{Employee}$$

$$\Big)$$

$$\Big)$$

Find the employees who earn more than their managers, returning badge, surname and salary of both the employee and the manager



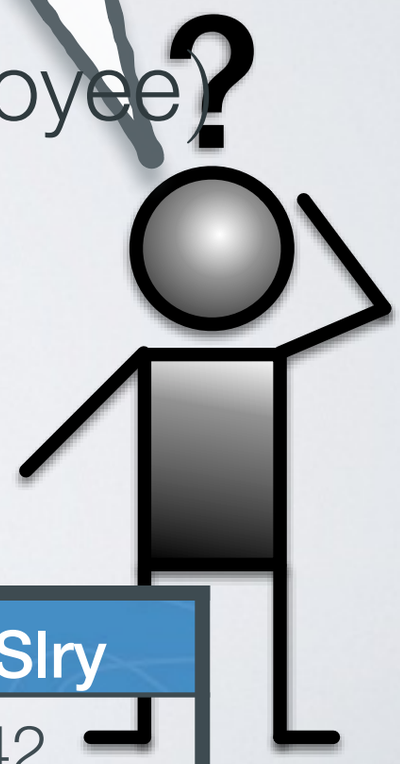
Badge	Surname	Salary	M.Bdg	M.Srn	M.Slry
7309	Rossi	45	5698	Bruni	42

© Stefano Marchesin

Relational Algebra: Exercise (6/8)

Find the employees who earn more than their managers, returning badge, surname and salary of both the employee and the manager

$$\pi_{\text{Badge,Surname,Salary,M.Bdg,M.Srn,M.Slry}}\Big(\rho_{\text{M.Bdg,M.Srn,M.Age,M.Slry}\leftarrow\text{Badge,Surname,Age,Salary}}(\text{Employee}) \bowtie_{\text{M.Bdg}=\text{Manager} \wedge \text{Salary}>\text{M.Slry}} \text{Manage} \bowtie_{\text{Employee}=\text{Badge}} \text{Employee}\Big)$$



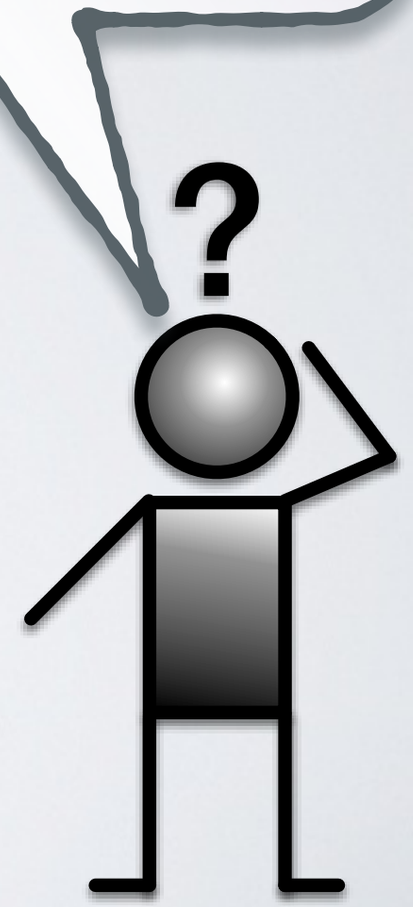
Badge	Surname	Salary	M.Bdg	M.Srn	M.Slry
7309	Rossi	45	5698	Bruni	42

Relational Algebra: Exercise (7/8)

Employee	Manager
7309	5698
5998	5698
9553	4076
5698	4076
4076	8123

Manage

Find the badge number of the managers whose employees **ALL** earn more than 42

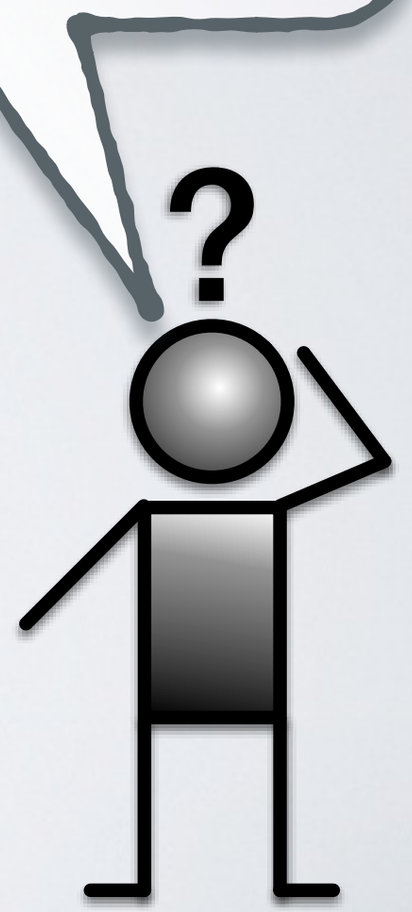


Employee

Badge	Surname	Age	Salary
7309	Rossi	34	45
5998	Bianchi	37	38
9553	Neri	42	35
5698	Bruni	43	42
4076	Mori	45	50
8123	Lupi	46	60

Relational Algebra: Exercise (7/8)

Find the badge number of the managers whose employees **ALL** earn more than 42



$$\pi_{\text{Manager}}(\text{Manage}) - \pi_{\text{Manager}}(\text{Manage} \bowtie_{\text{Employee}=\text{Badge}} \sigma_{\text{Salary}\leq 42}(\text{Employee}))$$

Manager
8123

Relational Algebra: Exercise (8/8)

Employee	Manager
7309	5698
5998	5698
9553	4076
5698	4076
4076	8123

Manage

Find the badge number of the managers who, in turn, have a manager

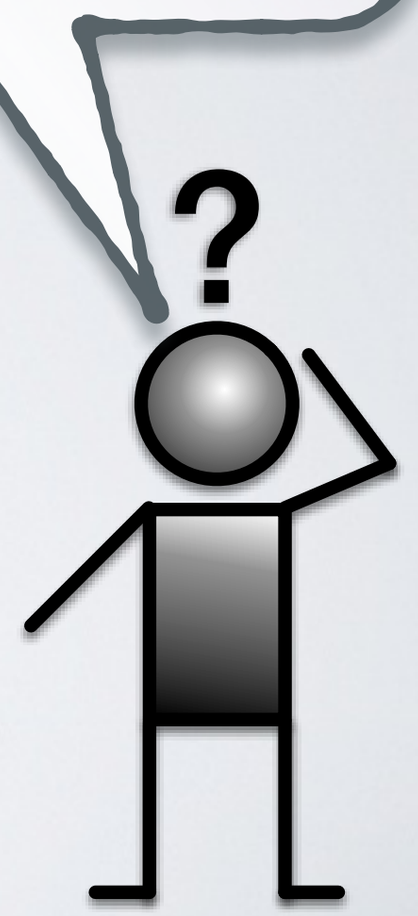


Employee

Badge	Surname	Age	Salary
7309	Rossi	34	45
5998	Bianchi	37	38
9553	Neri	42	35
5698	Bruni	43	42
4076	Mori	45	50
8123	Lupi	46	60

Relational Algebra: Exercise (8/8)

Find the badge number of the managers who, in turn, have a manager



$$\pi_{\text{WithMgr}}\Big(\text{Manage} \bowtie_{\text{Manager}=\text{WithMgr}} \rho_{\text{WithMgr}\leftarrow\text{Employee}}(\text{Manage})\Big)$$

WithMgr
5698
4076

Relational Algebra: Exercise (8/8)

$$\rho_{\text{WithMgr} \leftarrow \text{Manager}}\Big(\pi_{\text{Manager}}(\text{Manage})\Big)$$

$$\cap$$

$$\rho_{\text{WithMgr} \leftarrow \text{Employee}}\Big(\pi_{\text{Employee}}(\text{Manager})\Big)$$

WithMgr
5698
4076

Find the badge number of the managers who, in turn, have a manager



Relational Algebra: Exercise (8/8)

Find the badge number of the managers who, in turn, have a manager

$$\sigma_{\text{Manager}=\text{Employee}}(\text{Manage})$$

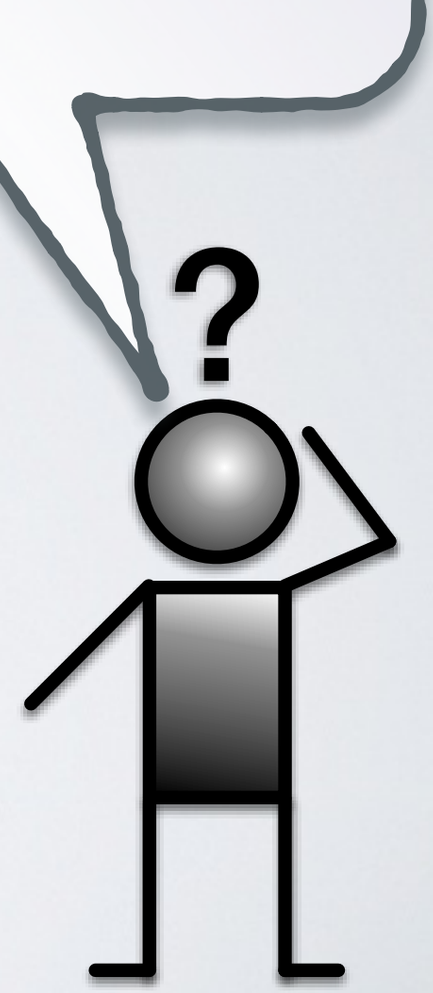


Relational Algebra: Exercise (8/8)

Find the badge number of the managers who, in turn, have a manager

$\sigma_{Manager=Employee}(Manage)$

This is searching for tuples where in the **SAME** tuple Employee and Manager are equal



© Stefano Marchesin

Issues with NULL Values in Relational Algebra

Employee

Badge	Surname	Age	Branch
7309	Rossi	34	Roma
5998	Bianchi	42	Milano
9553	Neri	NULL	Milano

Issues with NULL Values in Relational Algebra

Employee

Badge	Surname	Age	Branch
7309	Rossi	34	Roma
5998	Bianchi	42	Milano
9553	Neri	NULL	Milano

$$\sigma_{\text{Age}>40}(\text{Employee})$$

Badge	Surname	Age	Branch
5998	Bianchi	42	Milano

Issues with NULL Values in Relational Algebra

Employee

Badge	Surname	Age	Branch
7309	Rossi	34	Roma
5998	Bianchi	42	Milano
9553	Neri	NULL	Milano

$$\sigma_{\text{Age}>40}(\text{Employee}) \cup \sigma_{\text{Age}\leq 40}(\text{Employee}) \neq \text{Employee}$$

Selections are evaluated separately

Issues with NULL Values in Relational Algebra

Employee

Badge	Surname	Age	Branch
7309	Rossi	34	Roma
5998	Bianchi	42	Milano
9553	Neri	NULL	Milano

$$\sigma_{\text{Age}>40 \ \vee \ \text{Age}\leq 40}(\text{Employee}) \neq \text{Impiegato}$$

Atomic conditions are evaluated separately

Managing the NULL Values

- To manage NULL values we introduce specific conditions:

  IS NULL

  IS NOT NULL

Employee

Badge	Surname	Age	Branch
7309	Rossi	34	Roma
5998	Bianchi	42	Milano
9553	Neri	NULL	Milano

σ_{Age} **IS NULL** (Employee)

Badge	Surname	Age	Branch
9553	Neri	NULL	Milano

σ_{Age} **IS NOT NULL** (Employee)

Badge	Surname	Age	Branch
7309	Rossi	34	Roma
5998	Bianchi	42	Milano

Managing the NULL Values: Example

Employee

Badge	Surname	Age	Branch
7309	Rossi	34	Roma
5998	Bianchi	42	Milano
9553	Neri	NULL	Milano

$$\sigma_{\text{Age}>40 \,\vee\, \text{Age}\leq 40 \,\vee\, \text{Age } \textbf{IS NULL}}(\text{Employee}) = \text{Employee}$$

$$\left.\begin{array}{l} \sigma_{\text{Age}>40}(\text{Employee}) \\ \cup \\ \sigma_{\text{Age}\leq 40}(\text{Employee}) \\ \cup \\ \sigma_{\text{Age } \textbf{IS NULL}}(\text{Employee}) \end{array}\right\} = \text{Employee}$$

Join: Dandling Tuples

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Dep	Manage
A	Mori
B	Bruni
D	Verdi

Employee ⋈ Department

Surname	Dep	Manager
Rossi	A	Mori
Neri	B	Bruni
Bianchi	B	Bruni

- Some tuples are dandling and do not contribute to the results

Full Outer Join

Given two relations $R_1(XW)$ and $R_2(YZ)$ with W and Z sets of attributes on the same domain, the full outer join is

$$R_1 ⟗_{W\ \theta\ Z} R_2 = R_1 ⋈_{W\ \theta\ Z} R_2$$
$$\bigcup$$
$$\left(R_1 - \pi_{X,W}(R_1 ⋈_{W\ \theta\ Z} R_2)\right) \times \{Y = \text{NULL}, Z = \text{NULL}\}$$
$$\bigcup$$
$$\left(R_2 - \pi_{Y,Z}(R_1 ⋈_{W\ \theta\ Z} R_2)\right) \times \{X = \text{NULL}, W = \text{NULL}\}$$

- The full outer join extends, with NULL values, the tuples which are excluded from the theta-join, also called **inner join**, keeping tuples from both operands

Left Outer Join

Given two relations $R_1(XW)$ and $R_2(YZ)$ with W and Z sets of attributes on the same domain, the left outer join is

$$R_1 ⟕_{W\ \theta\ Z} R_2 = R_1 \bowtie_{W\ \theta\ Z} R_2$$
$$\bigcup$$
$$\left(R_1 - \pi_{X,W}(R_1 \bowtie_{W\ \theta\ Z} R_2)\right) \times \{Y = \text{NULL}, Z = \text{NULL}\}$$

Right Outer Join

Given two relations $R_1(XW)$ and $R_2(YZ)$ with W and Z sets of attributes on the same domain, the right outer join is

$$R_1 \overset{\sqsubset}{\bowtie}_{W\ \theta\ Z} R_2 = R_1 \bowtie_{W\ \theta\ Z} R_2 \cup \left(R_2 - \pi_{Y,Z}(R_1 \bowtie_{W\ \theta\ Z} R_2)\right) \times \{X = \text{NULL}, W = \text{NULL}\}$$

Full Outer Join: Example

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Code	Manager
A	Mori
B	Bruni
D	Verdi

Employee $\bowtie_{Dep=Code}$ Department

Surname	Dep	Code	Manager
Rossi	A	A	Mori
Neri	B	B	Bruni
Bianchi	B	B	Bruni
Neri	C	NULL	NULL
NULL	NULL	D	Verdi

Full Outer Join: Example

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Code	Manager
A	Mori
B	Bruni
D	Verdi

Employee $\bowtie_{\text{Dep}=\text{Code}}$ **Department**

Employee $\bowtie_{\text{Dep}=\text{Code}}$ **Department**

Surname	Dep	Code	Manager
Rossi	A	A	Mori
Neri	B	B	Bruni
Bianchi	B	B	Bruni
Neri	C	NULL	NULL
NULL	NULL	D	Verdi

Full Outer Join: Example

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Code	Manager
A	Mori
B	Bruni
D	Verdi

$$(\textbf{Employee} - \pi_{\text{Surname,Dep}}(\textbf{Employee} \bowtie_{\text{Dep}=\text{Code}} \textbf{Department})) \times \{\text{Code} = \text{NULL}, \text{Manager} = \text{NULL}\}$$

Employee $\bowtie_{\text{Dep}=\text{Code}}$ Department

Surname	Dep	Code	Manager
Rossi	A	A	Mori
Neri	B	B	Bruni
Bianchi	B	B	Bruni
Neri	C	NULL	NULL
NULL	NULL	D	Verdi

Full Outer Join: Example

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Code	Manager
A	Mori
B	Bruni
D	Verdi

$$\left(\textbf{Department} - \pi_{\text{Code,Manager}}\left(\textbf{Employee} \bowtie_{\text{Rep}=\text{Code}} \textbf{Department}\right)\right) \times \{\text{Surname} = \text{NULL}, \text{Dep} = \text{NULL}\}$$

Employee $\bowtie_{\text{Dep}=\text{Code}}$ Department

Surname	Dep	Code	Manager
Rossi	A	A	Mori
Neri	B	B	Bruni
Bianchi	B	B	Bruni
Neri	C	NULL	NULL
NULL	NULL	D	Verdi

Left Outer Join: Example

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Code	Manager
A	Mori
B	Bruni
D	Verdi

Employee $\bowtie_{Dep=Code}$ Department

Surname	Dep	Code	Manager
Rossi	A	A	Mori
Neri	B	B	Bruni
Bianchi	B	B	Bruni
Neri	C	NULL	NULL

Left Outer Join: Example

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Code	Manager
A	Mori
B	Bruni
D	Verdi

Employee $\bowtie_{\text{Dep}=\text{Code}}$ **Department**

Employee $\bowtie_{\text{Dep}=\text{Code}}$ **Department**

Surname	Dep	Code	Manager
Rossi	A	A	Mori
Neri	B	B	Bruni
Bianchi	B	B	Bruni
Neri	C	NULL	NULL

Left Outer Join: Example

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Code	Manager
A	Mori
B	Bruni
D	Verdi

$$\left(\textbf{Employee} - \pi_{\text{Surname,Dep}}\left(\textbf{Employee} \bowtie_{\text{Dep}=\text{Code}} \textbf{Department}\right)\right) \times \{\text{Code} = \text{NULL}, \text{Manager} = \text{NULL}\}$$

Employee $\bowtie_{\text{Dep}=\text{Code}}$ Department

Surname	Dep	Code	Manager
Rossi	A	A	Mori
Neri	B	B	Bruni
Bianchi	B	B	Bruni
Neri	C	NULL	NULL

Right Outer Join: Example

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Code	Manager
A	Mori
B	Bruni
D	Verdi

Employee $\bowtie_{Dep=Code}$ Department

Surname	Dep	Code	Manager
Rossi	A	A	Mori
Neri	B	B	Bruni
Bianchi	B	B	Bruni
NULL	NULL	D	Verdi

Right Outer Join: Example

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Code	Manager
A	Mori
B	Bruni
D	Verdi

Employee $\bowtie_{\text{Dep}=\text{Code}}$ **Department**

Employee $\bowtie_{\text{Dep}=\text{Code}}$ **Department**

Surname	Dep	Code	Manager
Rossi	A	A	Mori
Neri	B	B	Bruni
Bianchi	B	B	Bruni
NULL	NULL	D	Verdi

Right Outer Join: Example

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	C

Department (R_2)

Code	Manager
A	Mori
B	Bruni
D	Verdi

$$\left(\textbf{Department} - \pi_{\text{Code,Manager}}\left(\textbf{Employee} \bowtie_{\text{Rep}=\text{Code}} \textbf{Department}\right)\right) \times \{\text{Surname} = \text{NULL}, \text{Dep} = \text{NULL}\}$$

Employee $\bowtie_{\text{Dep}=\text{Code}}$ Department

Surname	Dep	Code	Manager
Rossi	A	A	Mori
Neri	B	B	Bruni
Bianchi	B	B	Bruni
NULL	NULL	D	Verdi

© Stefano Marchesin

Outer Join and NULL Values

Employee (R_1)

Surname	Dep
Rossi	A
Neri	B
Bianchi	B
Neri	NULL

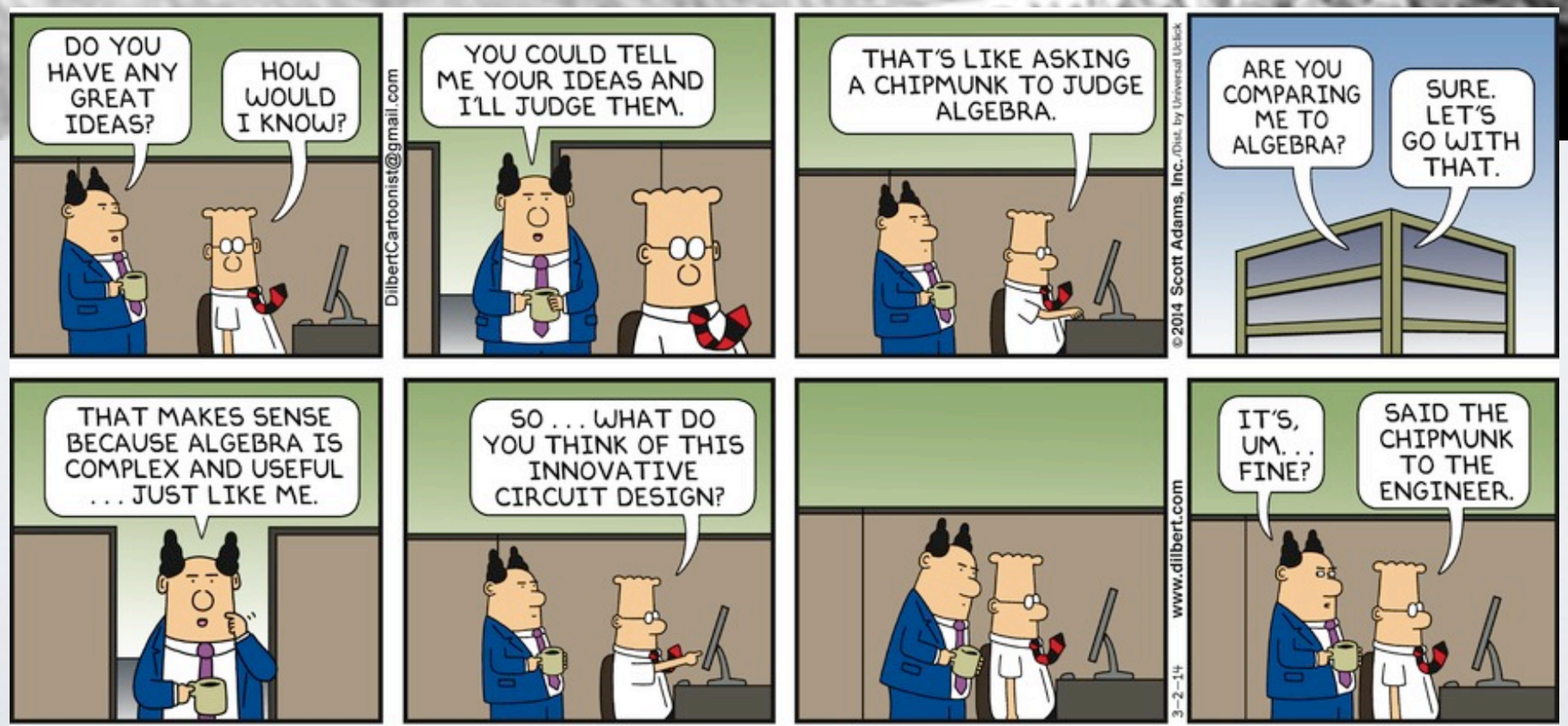
Department (R_2)

Code	Manager
A	Mori
B	Bruni
NULL	Verdi

Employee $\bowtie_{Dep=Code}$ Department

Surname	Dep	Code	Manager
Rossi	A	A	Mori
Neri	B	B	Bruni
Bianchi	B	B	Bruni
Neri	NULL	NULL	NULL
NULL	NULL	NULL	Verdi

Questions?


DO YOU HAVE ANY GREAT IDEAS?
HOW WOULD I KNOW?
DilbertCartoonist@gmail.com
YOU COULD TELL ME YOUR IDEAS AND I'LL JUDGE THEM.
THAT'S LIKE ASKING A CHIPMUNK TO JUDGE ALGEBRA.
ARE YOU COMPARING ME TO ALGEBRA?
SURE. LET'S GO WITH THAT.
© 2014 Scott Adams, Inc. /Dist. by Universal Uclick
THAT MAKES SENSE BECAUSE ALGEBRA IS COMPLEX AND USEFUL . . . JUST LIKE ME.
SO . . . WHAT DO YOU THINK OF THIS INNOVATIVE CIRCUIT DESIGN?
IT'S, UM. . . FINE?
SAID THE CHIPMUNK TO THE ENGINEER.
www.dilbert.com
3-2-14